Project 3 Architecture

Wednesday, March 18, 2026 6:02 PM

Recap:

RAG
↳ web page
↓
Recently launched web page
↓
Scraped content from webpage
↓
web base loader ← Document loader
↓
scrape
↓
content (a New Content).
↓
Split ⇒ Text Splitter
↳ chunks
↓
Chroma db ⇒ . from_documents (chunks) ← list
persist_directory
embedding
384 ← Sentence trans
1563 ← open AI
3072 ← Gemini
↓
Retrieve doc ⇒ relevant to query.
↓
vdb. as retriever (' ')
↓
4 doc.

↓

+ doc.

Content

Generation → Chain retriever | llm | str par

↓

Before info User · query.

'langchain - middle' → 100's

→ Youtube → rel video → web article

↓

X Youtube video link → Application → web article

1) Conv - Youtube video → transcript → api's

X Transcript

↓

langchain. doc. Loader

↓ Tool

↓ Youtube Loader

↓

Transcript

2) Give transcript to LLM → Detailed prompt to LLM.

↓

Summarize Transcript

↓ Article

↓

When the # transcript len.

But → A long YT video → # Tokens → large

↓

Tokens may not fit in LLM Context window.

↓

LLM cannot summarize whole transcript well

LLM

Content length { gpt-5, gemini 2.5.

↓

≈ 1 akh / 10,000 Tokens.

↓

problem

Content lengths are long

BUT effective Content length → 25% whole Content length

↓

length of input > 25%.

↓

perf of LLM response

↓

Low.

Recent Stat

↓

Tokens > 5000

↓

Quality of resp → Not good

YT video → length ≈ 30 min → # words → Transcript around 5000.

↓

How to solve?

↓

Text Splitter

Split text → 5000 tokens.

words | tokens > 5000

Split the doc chunks

Summarize chain()

Deprecated

Combine all the summaries

Sch1 ← Chunk1

Sch2 ← Chunk2

Sch3 ← Chunk3

Write a function to perform.

'Middle ware' LC 1.0

Adv: Interrupt the langchain agent & perform operation

Custom

Built-In → Summarizer

Summ - Text X

Summarize history of Conv.

Chatbot → Summarize history of Conv.

Chatbot

Aware of Context

Add: past conv to the memory.

Size memory inc

Input LLM

Input- LLM

↓

Token Inc

Context window

↳ limit → amt of past Conv

↓

Input to LLM.

Conv in memory → large Inc

↓

Summarize past Conv.

↓

Middle ware

Summarization

↓ token : 5000.

memory ← # 5000.

> 5000

Summary past Conv

+ nc1

+ nc2

+ nc3

⋮

Token > 5000.

Again it summarizes Conv.

Like a loop. { Middleware

↓

Summarize Conv